

Engineering Physics Lab

Engineering Physics Lab is a course that is designed to provide students with hands-on experience in the application of physics concepts in engineering. The course typically includes experiments that demonstrate the principles of mechanics, optics, electricity, magnetism, and thermodynamics. Here are some key points to consider when studying for an Engineering Physics Lab course:

1. **Preparation:** Before each lab session, it is important to read the lab manual and understand the objectives of the experiment. This will help you to perform the experiment more effectively and to understand the results.
2. **Data collection:** During the experiment, it is important to carefully record all measurements and observations. This will help you to analyze the data and draw conclusions.
3. **Data analysis:** After the experiment, you will need to analyze the data and compare it to the theoretical predictions. This will help you to understand the underlying physics principles and to identify any sources of error.
4. **Lab report:** A lab report is typically required for each experiment. The report should include a description of the experiment, the data collected, the analysis of the data, and the conclusions drawn.
5. **Safety:** Safety is always a top priority in any lab setting. Be sure to follow all safety guidelines and to wear appropriate protective equipment.

Studying for an Engineering Physics Lab course requires a combination of theoretical knowledge and practical skills. By carefully preparing for each experiment, collecting and analyzing data, and writing thorough lab reports, you can develop a deep understanding of the principles of physics and their application in engineering.

List of Experiments

1. **The Milgram Experiment:** This experiment was conducted by Stanley Milgram in 1961 to study obedience to authority figures. It involved participants administering electric shocks to a “learner” when they answered questions incorrectly.
2. **The Stanford Prison Experiment:** This experiment was conducted by Philip Zimbardo in 1971 to study the psychological effects of becoming a prisoner or prison guard. It involved college students being randomly assigned to play the role of either a prisoner or guard in a simulated prison environment.
3. **The Asch Conformity Experiment:** This experiment was conducted by Solomon Asch in 1951 to study the effects of group pressure on individual behavior. It involved participants being asked to match the length

of a line with comparison lines, while in a group where the majority gave incorrect answers.

4. **The Pavlov's Dog Experiment:** This experiment was conducted by Ivan Pavlov in the 1890s to study classical conditioning. It involved training a dog to associate the sound of a bell with food, causing the dog to salivate at the sound of the bell even when no food was present.
5. **The Bobo Doll Experiment:** This experiment was conducted by Albert Bandura in 1961 to study the effects of observing aggressive behavior. It involved children observing an adult model behaving aggressively towards a Bobo doll, and then being given the opportunity to interact with the doll themselves.

Any ten experiments (at least four from each group)

Here are ten experiments, with at least four from each group:

Group 1: Physics

1. **The Double-Slit Experiment:** This experiment demonstrates the wave-particle duality of light, where light behaves as both a wave and a particle.
2. **The Michelson-Morley Experiment:** This experiment was designed to detect the presence of aether, a hypothetical medium for the propagation of light. The results showed that aether did not exist.
3. **The Millikan Oil Drop Experiment:** This experiment measured the charge of an electron by balancing the gravitational and electric forces on tiny charged droplets of oil.
4. **The Rutherford Gold Foil Experiment:** This experiment showed that atoms have a small, dense, positively charged nucleus, surrounded by a cloud of negatively charged electrons.

Group 2: Chemistry

1. **The Iodine Clock Reaction:** This experiment demonstrates the concept of reaction rates and how they can be affected by factors such as concentration and temperature.
2. **The Briggs-Rauscher Oscillating Reaction:** This experiment is a visually striking example of a chemical oscillator, where the reaction mixture changes color periodically.
3. **The Belousov-Zhabotinsky Reaction:** This is another example of a chemical oscillator, where the reaction mixture exhibits spatiotemporal patterns.
4. **The Thermite Reaction:** This experiment demonstrates an exothermic redox reaction, where iron oxide reacts with aluminum to produce iron and aluminum oxide.

Group 3: Biology

1. **The Hershey-Chase Experiment:** This experiment showed that DNA, not protein, is the genetic material of phages, viruses that infect bacteria.
2. **The Meselson-Stahl Experiment:** This experiment demonstrated that DNA replication is semi-conservative, where each strand of the original DNA molecule serves as a template for the synthesis of a new strand.

Group A

- Group A is a term that can refer to many different things depending on the context.
- In the context of biology, Group A Streptococcus (GAS) is a type of bacteria that can cause a variety of infections.
- In the context of sports, Group A can refer to a specific group of teams or athletes in a tournament or competition.
- In the context of vitamins, Group A refers to fat-soluble vitamins, including vitamin A, D, E, and K.
- In the context of blood types, Group A refers to the A blood group, one of the four main blood groups in the ABO blood group system.
- In the context of automobiles, Group A can refer to a specific category of racing cars.
- In the context of education, Group A can refer to a specific group of students or subjects.
- In the context of business, Group A can refer to a specific group of companies or products.

1. To determine the wavelength of sodium light by Newton's ring experiment.

- Newton's ring experiment is a method to determine the wavelength of light using the phenomenon of interference.
- In this experiment, a plano-convex lens is placed on a flat glass plate, and a monochromatic light source, such as sodium light, is used to illuminate the setup.
- The light waves reflected from the top and bottom surfaces of the air film between the lens and the glass plate interfere with each other, producing a pattern of bright and dark rings known as Newton's rings.
- The radius of the rings can be measured using a traveling microscope, and the wavelength of the light can be calculated using the formula:
$$r_n^2 = \frac{D^2(n-m)^2}{4R}$$
 where r_n is the radius of the nth dark ring, D is the diameter of the lens, R is the radius of curvature of the lens, and m is the number of rings between the two diameters being measured.
- This experiment is named after Sir Isaac Newton, who first described the phenomenon of interference in his book "Opticks" in 1704.

2. To determine the wavelength of different spectral lines of mercury light using plane

- The wavelength of different spectral lines of mercury light can be determined using a plane transmission diffraction grating.
- A diffraction grating is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions.
- The directions of these beams depend on the spacing of the grating and the wavelength of the light.
- By measuring the angles of the diffracted beams, the wavelength of the light can be calculated using the grating equation.
- The grating equation is given by: $n\lambda = d(\sin i + \sin m)$, where n is the order of the diffraction, λ is the wavelength of the light, d is the grating spacing, i is the angle of incidence, and m is the angle of diffraction.
- To determine the wavelength of different spectral lines of mercury light, the light is passed through a diffraction grating and the angles of the diffracted beams are measured.
- The grating equation is then used to calculate the wavelength of the light for each spectral line.
- This experiment can be performed using a spectrometer, which is an instrument used to measure the properties of light over a specific portion of the electromagnetic spectrum.
- The spectrometer consists of a collimator, a diffraction grating, and a telescope.
- The collimator produces a parallel beam of light, which is diffracted by the grating.
- The telescope is used to measure the angles of the diffracted beams.
- By performing this experiment, the wavelength of different spectral lines of mercury light can be accurately determined.

Transmission Grating

A transmission grating is a type of diffraction grating that is used in transmission. This is in contrast to the more common reflection gratings, which reflect all incident light, as far as it is not absorbed or scattered.

Transmission gratings are designed to disperse incident light at specific angles. Angular dispersion is a function of angle of incidence and groove spacing. Dispersion increases as angle of incidence increases or as groove spacing decreases.

Transmission gratings are manufactured in the same way as ruled reflection gratings in which a replica of a master grating is created by duplicating its grooved surface with a thin, vacuum deposition-coated layer, except they have no reflective overcoat.

Transmission gratings offer low polarization sensitivity as opposed to reflection gratings because incident light is not reflected by a mirror coating such as aluminum. Since light is transmitted through the grating, transmission gratings can be used in compact, in-line configurations.

Transmission gratings can be volume Bragg gratings or surface relief gratings.

3. To determine the specific rotation of cane sugar solution using polarimeter.

A polarimeter is an instrument used to measure the angle of rotation of plane-polarized light as it passes through an optically active substance. The specific rotation of a substance is defined as the angle of rotation of plane-polarized light per unit path length and per unit concentration of the substance.

To determine the specific rotation of cane sugar solution using a polarimeter, the following steps can be followed:

1. Prepare a solution of known concentration of cane sugar by dissolving a known mass of cane sugar in a known volume of water.
2. Fill a polarimeter tube with the cane sugar solution and place it in the polarimeter.
3. Adjust the analyzer of the polarimeter until the field of view is dark.
4. Record the angle of rotation shown on the polarimeter scale.
5. Repeat the above steps for different concentrations of cane sugar solution.
6. Calculate the specific rotation of cane sugar solution using the formula:
specific rotation = angle of rotation / (path length x concentration).

By following these steps, the specific rotation of cane sugar solution can be determined using a polarimeter. This value can be used to determine the concentration of cane sugar in unknown solutions by measuring their angle of rotation and using the specific rotation to calculate their concentration.

4. To determine the focal length of the combination of two lenses separated by a distance

The focal length of the combination of two lenses separated by a distance can be determined using the formula:

$$1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

Where: - f is the focal length of the combination of the two lenses - f₁ is the focal length of the first lens - f₂ is the focal length of the second lens - d is the distance between the two lenses

To determine the focal length of the combination of the two lenses, the following steps can be followed: 1. Measure the focal length of the first lens (f₁) using a suitable method. 2. Measure the focal length of the second lens (f₂) using a suitable method. 3. Measure the distance between the two lenses (d). 4. Substitute the values of f₁, f₂, and d into the formula above to calculate the focal length of the combination of the two lenses (f).

It is important to note that the formula above is only valid when the two lenses are thin and the distance between them is small compared to their focal lengths. If the lenses are thick or the distance between them is large, more complex calculations may be required to determine the focal length of the combination of the two lenses.

Focal Length of Combination of Lenses

The formula for the focal length of a combination of lenses is given by the formula:

$$1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

Where: - f is the focal length of the combination of lenses - f_1 is the focal length of the first lens - f_2 is the focal length of the second lens - d is the distance between the two lenses

This formula can be derived using the thin lens formula and the principle of superposition. The thin lens formula states that for a thin lens, the distance of the object, image, and focal length are related by the equation:

$$1/f = 1/u + 1/v$$

Where: - f is the focal length of the lens - u is the distance of the object from the lens - v is the distance of the image from the lens

When two lenses are placed close to each other, the image formed by the first lens acts as the object for the second lens. Using the principle of superposition, the total power of the combination of lenses is the sum of the powers of the individual lenses. The power of a lens is given by the reciprocal of its focal length. Therefore, the total power of the combination of lenses is given by:

$$P = P_1 + P_2 = 1/f_1 + 1/f_2$$

Substituting the value of P in the thin lens formula for the combination of lenses, we get:

$$1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

This is the formula for the focal length of a combination of lenses. It can be used to calculate the focal length of a combination of two thin lenses placed close to each other. The distance between the lenses should be small compared to the focal lengths of the lenses for the formula to be valid.

5. To measure attenuation in an optical fiber.

Attenuation in an optical fiber refers to the reduction in the intensity of light as it travels through the fiber. This reduction in intensity is due to absorption, scattering, and other loss mechanisms. Measuring attenuation in an optical fiber is important for determining the performance of the fiber and for identifying any issues that may affect its performance.

Here are the steps to measure attenuation in an optical fiber:

1. Set up a light source and a power meter: A light source is used to inject light into the fiber, and a power meter is used to measure the power of the light exiting the fiber.

2. Measure the input power: The input power is the power of the light entering the fiber. This is measured by connecting the light source directly to the power meter and recording the power reading.
3. Measure the output power: The output power is the power of the light exiting the fiber. This is measured by connecting the fiber to the light source and the power meter and recording the power reading.
4. Calculate the attenuation: The attenuation is calculated by subtracting the output power from the input power and dividing by the length of the fiber. The result is expressed in decibels per kilometer (dB/km).
5. Repeat the measurement: To ensure accuracy, the measurement should be repeated several times and the results averaged.

By following these steps, one can accurately measure the attenuation in an optical fiber. This information can be used to assess the performance of the fiber and to identify any issues that may affect its performance.

6. To determine the wavelength of He-Ne laser light using single slit diffraction.

1. The experiment is set up by shining a He-Ne laser light through a single slit and onto a screen.
2. The diffraction pattern produced on the screen is observed and measured.
3. The distance between the central maximum and the first minimum on either side is measured.
4. The slit width and the distance between the slit and the screen are also measured.
5. Using the formula for single slit diffraction, the wavelength of the He-Ne laser light can be calculated.
6. The formula for single slit diffraction is: $\lambda = (a \cdot \sin \theta) / m$, where λ is the wavelength, a is the slit width, θ is the angle of diffraction, and m is the order of the minimum.
7. The angle of diffraction can be calculated using the distance between the central maximum and the first minimum, and the distance between the slit and the screen.
8. By substituting the measured values into the formula, the wavelength of the He-Ne laser light can be determined.

7. To study the polarization of light using He-Ne laser light.

- Polarization refers to the orientation of the oscillations of an electromagnetic wave, such as light.
- Light can be polarized by passing it through a polarizing filter, which only allows light waves oscillating in a specific direction to pass through.
- A He-Ne laser emits light that is already polarized, due to the nature of the laser's construction and operation.

- The polarization of the He-Ne laser light can be studied by passing it through a polarizing filter and observing the intensity of the transmitted light.
- By rotating the polarizing filter, the transmitted intensity will vary, reaching a maximum when the filter's transmission axis is aligned with the polarization direction of the laser light, and a minimum when the transmission axis is perpendicular to the polarization direction.
- This experiment can be used to demonstrate the concept of polarization and to measure the polarization direction of the He-Ne laser light.

8. To determine the wavelength of sodium light with the help of Fresnel's bi-prism.

Fresnel's bi-prism is an optical instrument used to determine the wavelength of light. It consists of two prisms placed base to base, forming a bi-prism. When a beam of light passes through the bi-prism, it is split into two beams, which interfere to produce a pattern of fringes.

To determine the wavelength of sodium light using Fresnel's bi-prism, the following steps can be followed:

1. Set up the apparatus, consisting of a sodium lamp, a collimator, a bi-prism, and a telescope.
2. Adjust the collimator to produce a parallel beam of light.
3. Place the bi-prism in the path of the light beam, such that the light is split into two beams.
4. Observe the interference pattern produced by the two beams using the telescope.
5. Measure the distance between the fringes and the distance between the slit and the screen.
6. Use the formula for the wavelength of light in terms of the fringe spacing, the distance between the slit and the screen, and the angle of the prism to calculate the wavelength of the sodium light.

This experiment can be used to accurately determine the wavelength of sodium light, which is approximately 589 nanometers. It is an important experiment in the field of optics and is commonly performed in physics laboratories.

9. To determine the coefficient of viscosity of a given liquid.

Viscosity is a measure of a fluid's resistance to flow. The coefficient of viscosity is a property of a fluid that determines the amount of internal resistance to flow. There are several methods to determine the coefficient of viscosity of a given liquid, including:

1. **Ostwald viscometer:** This method involves measuring the time it takes for a fixed volume of liquid to flow through a capillary tube under the

influence of gravity. The coefficient of viscosity can be calculated using Poiseuille's equation.

2. **Falling sphere viscometer:** This method involves measuring the terminal velocity of a sphere falling through a liquid. The coefficient of viscosity can be calculated using Stokes' law.
3. **Rotational viscometer:** This method involves measuring the torque required to rotate a spindle at a constant speed in a liquid. The coefficient of viscosity can be calculated using the relationship between torque, spindle speed, and the dimensions of the spindle and liquid container.

Each of these methods has its advantages and limitations, and the choice of method depends on the specific properties of the liquid being tested and the accuracy required. It is important to carefully calibrate the equipment and follow standard procedures to obtain accurate results.

10. To determine the value of acceleration due to gravity (g) using compound pendulum.

The compound pendulum is a rigid body swinging in a vertical plane about a horizontal axis passing through the body. The time period of a compound pendulum is given by the formula $T = 2\pi\sqrt{I/mgd}$, where I is the moment of inertia of the pendulum about the axis of rotation, m is the mass of the pendulum, g is the acceleration due to gravity, and d is the distance between the center of mass of the pendulum and the axis of rotation.

To determine the value of g using a compound pendulum, the following steps can be followed:

1. Suspend the compound pendulum from a rigid support and set it into oscillation.
2. Measure the time period of the pendulum for a given number of oscillations using a stopwatch.
3. Calculate the average time period of the pendulum.
4. Measure the mass of the pendulum using a balance.
5. Determine the distance between the center of mass of the pendulum and the axis of rotation.
6. Calculate the moment of inertia of the pendulum about the axis of rotation using the parallel axis theorem.
7. Substitute the values of the time period, mass, distance, and moment of inertia in the formula $T = 2\pi\sqrt{I/mgd}$ and solve for g .

This experiment can be repeated with different pendulums to obtain more accurate results. The value of g obtained from this experiment can be compared with the standard value of g to check the accuracy of the experiment.

Group B

Group B refers to a set of regulations introduced in 1982 for competition vehicles in sportscar racing and rallying regulated by the Fédération Internationale de l'Automobile (FIA). The Group B regulations fostered some of the quickest, most powerful, and most sophisticated rally cars ever built and is commonly referred to as the golden era of rallying. However, the cost of competing in Group B, combined with a series of fatal accidents, eventually led to the cancellation of the Group B category at the end of the 1986 season.

Some key points about Group B include: - Introduced in 1982 for competition vehicles in sportscar racing and rallying. - Fostered some of the quickest, most powerful, and most sophisticated rally cars ever built. - Commonly referred to as the golden era of rallying. - The cost of competing in Group B, combined with a series of fatal accidents, led to its cancellation at the end of the 1986 season.

1. To determine the energy band gap of a given semiconductor material.

- The energy band gap of a semiconductor material is the difference in energy between the top of the valence band and the bottom of the conduction band.
- The band gap is a crucial property of a semiconductor material, as it determines the material's electrical conductivity and optical properties.
- There are several methods to determine the energy band gap of a semiconductor material, including optical absorption, photoluminescence, and electrical measurements.
- In the optical absorption method, the absorption coefficient of the material is measured as a function of the photon energy. The absorption edge, where the absorption coefficient starts to increase rapidly, corresponds to the energy band gap of the material.
- In the photoluminescence method, the material is excited with light of a known energy, and the emitted light is analyzed to determine the energy band gap.
- In electrical measurements, the current-voltage characteristics of the material are measured at different temperatures, and the energy band gap is determined from the temperature dependence of the current.
- The choice of method depends on the specific properties of the material and the experimental conditions.

2. To study Hall effect and determine Hall coefficient, carrier density and mobility of a

The Hall effect is defined as the difference in voltage generated across a current-carrying conductor, transverse to an electrical current in the conductor and an applied magnetic field perpendicular to the current. The Hall effect measurement is one of the most important characterization techniques for electronic materials.

The drift velocity acquired in unit applied electric field is known as the mobility of the carrier and is denoted by μ_H and is also called Hall mobility. The Hall coefficient (RH) and mobility (μ_H) are independent of magnetic and electric fields, however they change with film thickness and composition of the film.

The Hall measurement, carried out in the presence of a magnetic field, yields the sheet carrier density n_s and the bulk carrier density n or p (for n-type or p-type material) if the conducting layer thickness of the sample is known. The variation of carrier concentration (n_H) depends on thickness and composition of Zn-Te films, observations show the carrier concentration decreases with increase of thickness while that increases with increase of at. wt.% of Zn in the film.

Given semiconductor material using Hall effect setup

The Hall effect is a phenomenon that occurs when a magnetic field is applied perpendicular to the flow of current in a conductor. This causes a voltage difference to develop across the conductor, perpendicular to both the current and the magnetic field. This voltage difference is known as the Hall voltage.

The Hall effect can be used to determine several properties of a given semiconductor material, including:

1. **Type of semiconductor:** The sign of the Hall voltage can be used to determine whether the semiconductor is a p-type or n-type material. A positive Hall voltage indicates that the material is p-type, while a negative Hall voltage indicates that it is n-type.
2. **Carrier concentration:** The magnitude of the Hall voltage is directly proportional to the carrier concentration in the semiconductor material. By measuring the Hall voltage, the carrier concentration can be determined.
3. **Mobility:** The mobility of the charge carriers in the semiconductor material can also be determined using the Hall effect. This is done by measuring the Hall voltage at different magnetic field strengths and using the relationship between the Hall voltage, magnetic field strength, and mobility.

To perform a Hall effect experiment on a given semiconductor material, a Hall effect setup is required. This typically consists of a sample of the semiconductor material, a source of current, a magnetic field, and a voltmeter to measure the Hall voltage. The sample is placed in the magnetic field, and a current is passed through it. The Hall voltage is then measured using the voltmeter.

In summary, the Hall effect can be used to determine several important properties of a given semiconductor material, including its type, carrier concentration, and mobility. This is done using a Hall effect setup, which typically consists of a sample of the material, a source of current, a magnetic field, and a voltmeter to measure the Hall voltage.

3. To determine the variation of magnetic field with the distance along the axis of a current

The magnetic field produced by a current-carrying wire can be determined using the Biot-Savart law. This law states that the magnetic field at a point is proportional to the current, the length of the wire element, and the inverse square of the distance from the wire element to the point.

To determine the variation of the magnetic field with distance along the axis of a current-carrying wire, the following steps can be followed:

1. Set up the experiment by placing a current-carrying wire along the axis of a magnetic field sensor.
2. Measure the magnetic field at different distances from the wire along the axis.
3. Plot the magnetic field strength against the distance from the wire.
4. Analyze the graph to determine the relationship between the magnetic field strength and the distance from the wire.

The graph should show that the magnetic field strength decreases as the distance from the wire increases. This relationship can be described mathematically using the inverse square law, which states that the magnetic field strength is inversely proportional to the square of the distance from the wire.

Carrying Coil and Estimate the Radius of the Coil

A carrying coil is a coil of wire that carries an electric current. The magnetic field produced by the current in the coil can be used to estimate the radius of the coil. Here are the steps to estimate the radius of the coil:

1. Measure the magnetic field strength at the center of the coil using a Gauss-meter.
2. Determine the number of turns in the coil.
3. Use the formula for the magnetic field at the center of a coil to calculate the radius of the coil. The formula is $B = (\mu_0 * I * N) / (2 * R)$, where B is the magnetic field strength, μ_0 is the permeability of free space, I is the current, N is the number of turns, and R is the radius of the coil.
4. Solve the equation for R to find the radius of the coil.

This method can be used to estimate the radius of a carrying coil. It is important to note that the accuracy of the estimate depends on the accuracy of the measurements and the assumptions made in the calculation.

4. To verify Stefan's law by electric method

Stefan's Law states that the total energy radiated per unit surface area of a black body across all wavelengths per unit time is directly proportional to the fourth power of the black body's thermodynamic temperature.

To verify Stefan's Law by electric method, the following steps can be followed:

1. Set up the apparatus, which includes a Leslie cube, a thermopile, a galvanometer, and a low voltage power supply.
2. Fill the Leslie cube with boiling water and place it on a heatproof mat.
3. Connect the thermopile to the galvanometer and the low voltage power supply.
4. Place the thermopile close to one of the faces of the Leslie cube.
5. Record the deflection of the galvanometer.
6. Repeat the experiment with the thermopile at different distances from the Leslie cube.
7. Plot a graph of the galvanometer deflection against the inverse square of the distance between the thermopile and the Leslie cube.
8. The graph should be a straight line, verifying Stefan's Law.

5. To determine resistance per unit length and specific resistance of a given resistance

1. Resistance per unit length is the resistance of a conductor divided by its length. It is usually measured in ohms per meter (Ω/m).
2. Specific resistance, also known as resistivity, is a measure of the intrinsic resistance of a material. It is usually measured in ohm-meters ($\Omega \cdot \text{m}$).
3. To determine the resistance per unit length of a given resistance, the resistance of the conductor is measured using an ohmmeter and then divided by its length.
4. To determine the specific resistance of a given resistance, the resistance per unit length is multiplied by the cross-sectional area of the conductor.
5. The specific resistance of a material is dependent on its temperature, so it is important to take this into account when making measurements.
6. The specific resistance of a material can also be determined using the formula $\rho = RA/l$, where ρ is the specific resistance, R is the resistance of the conductor, A is the cross-sectional area of the conductor, and l is the length of the conductor.
7. It is important to use accurate and precise measuring instruments when determining the resistance per unit length and specific resistance of a given resistance to ensure accurate results.

Using Carey Foster's Bridge

Carey Foster's Bridge is a method used to measure the resistance of a wire. It is an extension of the Wheatstone bridge, which is used to measure the resistance of an unknown resistor. Here are the steps to use Carey Foster's Bridge:

1. Connect the unknown resistance and the standard resistance in the two gaps of the bridge.
2. Adjust the sliding contact on the wire until the galvanometer shows zero deflection.
3. Measure the length of the wire between the two fixed points and the sliding contact.

4. Calculate the resistance of the wire using the formula: $R = (l_1 - l_2) * (\rho / L)$, where R is the resistance of the wire, l_1 and l_2 are the lengths of the wire between the fixed points and the sliding contact, ρ is the resistivity of the wire, and L is the total length of the wire.

Carey Foster's Bridge is a useful tool for accurately measuring the resistance of a wire. It is commonly used in laboratories and in the field of electrical engineering. It is important to use the bridge correctly and to take accurate measurements in order to obtain accurate results.

6. To study the resonance condition of a series LCR circuit.

A series LCR circuit consists of an inductor (L), a capacitor (C), and a resistor (R) connected in series. The resonance condition of a series LCR circuit occurs when the inductive reactance (X_L) equals the capacitive reactance (X_C). At resonance, the circuit behaves as a purely resistive circuit, and the current in the circuit is maximum.

The resonance condition can be mathematically expressed as:

$$X_L = X_C \quad 2\pi fL = 1 / (2\pi fC) \quad f = 1 / (2\pi \sqrt{LC})$$

Where f is the resonant frequency, L is the inductance, and C is the capacitance.

The resonance condition can be experimentally verified by measuring the current in the circuit as the frequency of the applied voltage is varied. At resonance, the current in the circuit will be maximum.

In summary, the resonance condition of a series LCR circuit occurs when the inductive reactance equals the capacitive reactance, resulting in a purely resistive circuit with maximum current. This condition can be mathematically expressed and experimentally verified.

7. To determine the electrochemical equivalent (ECE) of copper.

The electrochemical equivalent (ECE) of a substance is the mass of the substance deposited or dissolved at an electrode during electrolysis by the passage of one coulomb of electric charge. The ECE of copper can be determined by performing an experiment in which a known amount of electric charge is passed through a copper sulfate solution, and the mass of copper deposited on the cathode is measured.

The steps to determine the ECE of copper are as follows:

1. Set up the electrolysis apparatus with a copper sulfate solution and two copper electrodes.
2. Connect the electrodes to a power source and an ammeter to measure the current.
3. Record the initial mass of the cathode.

4. Pass a known amount of electric charge through the solution by running the current for a measured amount of time.
5. Record the final mass of the cathode.
6. Calculate the mass of copper deposited on the cathode by subtracting the initial mass from the final mass.
7. Divide the mass of copper deposited by the amount of electric charge passed to determine the ECE of copper.

The ECE of copper can be used to calculate the amount of copper that will be deposited during electrolysis under different conditions, such as varying the current or the time of electrolysis. It is an important parameter in electroplating and other industrial processes that involve the deposition of copper.

8. To calibrate the given ammeter and voltmeter by potentiometer.

Calibration is the process of comparing the readings of an instrument with a known standard to determine the accuracy of the instrument. In this experiment, we will calibrate an ammeter and a voltmeter using a potentiometer.

The following steps can be followed to calibrate the given ammeter and voltmeter by potentiometer:

1. Set up the circuit as shown in the diagram, with the ammeter and voltmeter connected in series with the potentiometer and a battery.
2. Adjust the potentiometer to obtain a known current through the circuit.
3. Record the reading of the ammeter and compare it with the known current.
4. If the ammeter reading is not equal to the known current, adjust the zero error of the ammeter accordingly.
5. Repeat the above steps for different values of current to obtain a calibration curve for the ammeter.
6. Similarly, adjust the potentiometer to obtain a known voltage across the voltmeter.
7. Record the reading of the voltmeter and compare it with the known voltage.
8. If the voltmeter reading is not equal to the known voltage, adjust the zero error of the voltmeter accordingly.
9. Repeat the above steps for different values of voltage to obtain a calibration curve for the voltmeter.

By following the above steps, the given ammeter and voltmeter can be calibrated using a potentiometer. It is important to calibrate these instruments to ensure accurate measurements in experiments.

Hysteresis (B-H curve) of a Specimen in the Form of a Transformer

Hysteresis is the phenomenon where the magnetic properties of a material depend on its magnetic history. The B-H curve, also known as the hysteresis loop, is a graphical representation of this phenomenon.

To draw the hysteresis loop of a specimen in the form of a transformer, the following steps can be followed:

1. Set up the transformer with the specimen as the core.
2. Apply an alternating current to the primary coil of the transformer.
3. Measure the magnetic field strength (H) and the magnetic flux density (B) of the specimen.
4. Plot the B-H curve by plotting the magnetic field strength (H) on the x-axis and the magnetic flux density (B) on the y-axis.
5. The resulting curve will show the hysteresis loop of the specimen.

The shape of the hysteresis loop can provide information about the magnetic properties of the specimen, such as its coercivity and remanence. These properties can be used to determine the suitability of the specimen for certain applications.

Determine its Hysteresis Loss

Hysteresis loss is the energy lost in the form of heat when a magnetic material is subjected to a changing magnetic field. It is a measure of the energy dissipated due to the magnetic hysteresis of the material. Here are the steps to determine the hysteresis loss of a magnetic material:

1. Obtain a sample of the magnetic material and measure its mass and volume.
2. Subject the sample to a changing magnetic field by placing it in a coil of wire carrying an alternating current.
3. Measure the magnetic field strength and the magnetic flux density of the sample at various points in the cycle.
4. Plot the magnetic field strength against the magnetic flux density to obtain the hysteresis loop of the material.
5. Calculate the area enclosed by the hysteresis loop. This area represents the energy lost per cycle due to hysteresis.
6. Multiply the area of the hysteresis loop by the frequency of the alternating current to obtain the hysteresis loss per unit time.
7. Divide the hysteresis loss per unit time by the volume of the sample to obtain the hysteresis loss per unit volume.

10. To measure high resistance by leakage method

The leakage method is a technique used to measure high resistance values. This method is based on the principle of measuring the leakage current flowing through the insulation of a conductor. The following are the steps involved in measuring high resistance by the leakage method:

1. The conductor whose resistance is to be measured is connected to a high voltage source.
2. The leakage current flowing through the insulation of the conductor is measured using a sensitive ammeter.
3. The resistance value is calculated by dividing the applied voltage by the measured leakage current.

It is important to note that the leakage method is only suitable for measuring high resistance values, as the leakage current is very small for low resistance values. Additionally, the accuracy of the measurement depends on the accuracy of the ammeter used to measure the leakage current.

Course outcomes:

1. Understanding of the subject matter: Students will gain a thorough understanding of the course material and be able to apply the concepts and principles learned in the course to real-world situations.
2. Development of critical thinking skills: Students will learn to analyze, evaluate, and synthesize information to make informed decisions and solve problems.
3. Improvement of communication skills: Students will learn to effectively communicate their ideas and thoughts through written and oral presentations.
4. Enhancement of teamwork and collaboration skills: Students will learn to work effectively in groups and collaborate with others to achieve common goals.
5. Acquisition of practical skills: Students will gain hands-on experience and develop practical skills relevant to their field of study.
6. Preparation for further study or career: Students will be prepared for further study or entry into the workforce in their chosen field.
7. Personal growth and development: Students will have the opportunity to explore their interests, develop their strengths, and grow as individuals.

At the end of the course, students will be able to:

1. Demonstrate a comprehensive understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.

4. Communicate their ideas and arguments clearly and effectively in written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in academic and professional settings.
7. Use technology effectively to access, organize, and present information.
8. Engage in lifelong learning and professional development.

CO CO Statement Bloom's

CO (Course Outcome) and CO (Course Objective) statements are used to describe the expected learning outcomes of a course. These statements are typically written in terms of what the student should be able to do or understand by the end of the course.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to help develop CO and CO statements by providing a structure for defining and organizing learning outcomes.

The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** recalling information from memory.
2. **Understanding:** comprehending the meaning of information.
3. **Applying:** using information in a new situation.
4. **Analyzing:** breaking down information into parts and understanding their relationships.
5. **Evaluating:** making judgments about the value of information.
6. **Creating:** combining information to form a new whole.

By using Bloom's Taxonomy to develop CO and CO statements, educators can ensure that their courses are designed to promote higher-order thinking skills and provide a well-rounded learning experience for their students.

Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, with an air bubble inside.
- When the level is placed on a surface, the bubble will move to the highest point of the tube.
- If the bubble is centered between two marks on the tube, the surface is level.
- Levels come in various sizes and shapes, including spirit levels, laser levels, and digital levels.
- They are commonly used in construction, carpentry, and surveying.
- Levels can also be used to hang pictures or shelves, or to level appliances such as washing machines or refrigerators.

- It is important to use a level when installing floors, cabinets, or countertops to ensure that they are even and stable.
- A level can also be used to check the slope of a surface, such as a driveway or a patio, to ensure proper drainage.
- Using a level can help to improve the accuracy and quality of a project.

CO-1 Apply the principle of interference and diffraction to find the wavelength of

Interference and diffraction are two phenomena that can be used to determine the wavelength of light. Here are the steps to apply these principles:

1. **Interference:** Interference occurs when two or more waves overlap and combine to form a new wave. This can be observed when light passes through a narrow slit or when two light sources are close together. The resulting pattern of light and dark bands is called an interference pattern.
2. **Young's Double Slit Experiment:** One way to observe interference is through Young's double slit experiment. In this experiment, light from a single source is passed through two narrow slits, creating two coherent light sources. The light from these sources interferes to produce an interference pattern on a screen.
3. **Calculating Wavelength:** The distance between the bright bands in the interference pattern is related to the wavelength of the light. By measuring this distance and knowing the distance between the slits and the screen, the wavelength of the light can be calculated using the formula: $\lambda = (d * x) / L$, where λ is the wavelength, d is the distance between the slits, x is the distance between the bright bands, and L is the distance between the slits and the screen.
4. **Diffraction:** Diffraction is the bending of waves around obstacles or through openings. This can be observed when light passes through a narrow slit or a small hole. The resulting pattern of light and dark bands is called a diffraction pattern.
5. **Single Slit Diffraction:** One way to observe diffraction is through single slit diffraction. In this experiment, light from a single source is passed through a narrow slit, creating a diffraction pattern on a screen.
6. **Calculating Wavelength:** The width of the central bright band in the diffraction pattern is related to the wavelength of the light. By measuring this width and knowing the width of the slit and the distance between the slit and the screen, the wavelength of the light can be calculated using the formula: $\lambda = (w * x) / L$, where λ is the wavelength, w is the width of the slit, x is the width of the central bright band, and L is the distance between the slit and the screen.

These are the basic principles of interference and diffraction that can be used

to find the wavelength of light. By conducting experiments and making careful measurements, the wavelength of light can be accurately determined.

Monochromatic and Polychromatic Light

- **Monochromatic light** refers to light that consists of a single wavelength or color. This type of light is typically produced by lasers or filtered light sources.
- **Polychromatic light**, on the other hand, refers to light that consists of multiple wavelengths or colors. This type of light is typically produced by natural light sources such as the sun or by artificial light sources such as incandescent bulbs.
- Monochromatic light is often used in scientific experiments and measurements because it is easier to control and analyze than polychromatic light.
- Polychromatic light is more commonly found in everyday life and is used for a variety of purposes, including lighting and color displays.
- The difference between monochromatic and polychromatic light can be observed by looking at a rainbow, which is formed when white light (polychromatic light) is split into its component colors (monochromatic light) by a prism or water droplets.
- Monochromatic light can be used to produce interference and diffraction patterns, which can be used to study the properties of light and matter.
- Polychromatic light can be used to produce a wide range of colors and effects, making it useful for artistic and decorative purposes.
- Both monochromatic and polychromatic light have their advantages and disadvantages, and the choice between the two depends on the specific application and desired outcome.

CO-2 Compute and analyze various electrical and electronic properties of a given

1. **Resistance:** The property of a material to resist the flow of electric current. It is measured in ohms (Ω) and can be calculated using Ohm's Law, which states that the resistance is equal to the voltage divided by the current.
2. **Capacitance:** The ability of a material to store electric charge. It is measured in farads (F) and is determined by the distance between the plates of a capacitor, the area of the plates, and the dielectric constant of the material between the plates.
3. **Inductance:** The property of a coil of wire to resist changes in electric current flowing through it. It is measured in henries (H) and is determined by the number of turns in the coil, the cross-sectional area of the coil, and the magnetic permeability of the core material.

4. **Impedance:** The total opposition to the flow of alternating current (AC) in an electric circuit. It is measured in ohms (Ω) and is a combination of resistance and reactance, which is the opposition to the flow of AC due to capacitance and inductance.
5. **Conductance:** The ability of a material to conduct electric current. It is the reciprocal of resistance and is measured in siemens (S).
6. **Transconductance:** The ratio of the change in the output current of a device to the change in the input voltage. It is measured in siemens (S) and is commonly used to characterize the performance of field-effect transistors (FETs).
7. **Permittivity:** The ability of a material to polarize in response to an applied electric field. It is measured in farads per meter (F/m) and is a measure of the material's ability to store electric charge.
8. **Permeability:** The ability of a material to support the formation of a magnetic field within itself. It is measured in henries per meter (H/m) and is a measure of the material's ability to conduct a magnetic field.

Material Analysis Using Various Experiments

Material analysis is the process of determining the chemical composition, structure, and properties of a material. This can be done through various experiments, including:

1. **Spectroscopy:** This technique involves analyzing the interaction between matter and electromagnetic radiation to determine the chemical composition of a material.
2. **X-ray diffraction:** This technique involves shining X-rays onto a material and analyzing the resulting diffraction pattern to determine the crystal structure of the material.
3. **Thermal analysis:** This technique involves measuring the physical and chemical changes that occur in a material as it is heated or cooled.
4. **Microscopy:** This technique involves using a microscope to observe the microstructure of a material.
5. **Mechanical testing:** This technique involves subjecting a material to various mechanical stresses to determine its mechanical properties, such as strength and ductility.

These are just a few examples of the many experiments that can be used to analyze materials. The specific experiment used will depend on the material being analyzed and the information that is being sought.

CO-3 Verify different established laws with the help of optical and electrical

There are several established laws in the field of optics and electricity that can be verified through experiments and observations. Some of these laws include:

1. **Ohm's Law:** This law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided the temperature remains constant. This law can be verified by measuring the current flowing through a conductor at different voltage levels and plotting a graph of voltage against current.
2. **Snell's Law:** This law describes the relationship between the angles of incidence and refraction of light when it passes from one medium to another. This law can be verified by measuring the angles of incidence and refraction of a light beam as it passes through different media and comparing the results with the predictions of Snell's Law.
3. **Kirchhoff's Laws:** These laws describe the conservation of charge and energy in electrical circuits. Kirchhoff's Current Law states that the sum of the currents entering a junction in a circuit must equal the sum of the currents leaving the junction. Kirchhoff's Voltage Law states that the sum of the voltage drops around any closed loop in a circuit must equal the sum of the voltage rises. These laws can be verified by measuring the currents and voltages in a circuit and checking that they satisfy the conditions of Kirchhoff's Laws.
4. **Lensmaker's Equation:** This equation describes the relationship between the focal length of a lens, the refractive indices of the media on either side of the lens, and the radii of curvature of the lens surfaces. This equation can be verified by measuring the focal length of a lens and comparing it with the value predicted by the Lensmaker's Equation.

These are just a few examples of the many established laws in the field of optics and electricity that can be verified through experiments and observations. By conducting such experiments, we can gain a deeper understanding of the fundamental principles that govern the behavior of light and electricity.

Experiments

An experiment is a scientific procedure undertaken to make a discovery, test a hypothesis, or demonstrate a known fact. Here are some key points to consider when conducting an experiment:

1. **Hypothesis:** A hypothesis is a proposed explanation for a phenomenon. It is important to have a clear and testable hypothesis before conducting an experiment.
2. **Variables:** Variables are factors that can be changed or controlled in an experiment. It is important to identify and control variables to ensure that the results of the experiment are accurate.

3. **Control group:** A control group is a group in an experiment that does not receive the treatment being tested. The control group is used as a baseline to compare the results of the treatment group.
4. **Data collection:** Data collection is the process of gathering and measuring information. It is important to have a clear and consistent method for collecting data to ensure that the results of the experiment are accurate.
5. **Data analysis:** Data analysis is the process of examining and interpreting the data collected during an experiment. It is important to use appropriate statistical methods to analyze the data and draw conclusions.
6. **Conclusion:** The conclusion is the final step in an experiment where the results are interpreted and the hypothesis is either supported or rejected. It is important to draw a clear and well-supported conclusion based on the data collected during the experiment.
7. **Replication:** Replication is the process of repeating an experiment to confirm the results. It is important to replicate experiments to ensure that the results are reliable and not due to chance.

These are some of the key points to consider when conducting an experiment. It is important to follow a rigorous scientific method to ensure that the results of the experiment are accurate and reliable.

Apply

1. The term “apply” can refer to the act of putting something into operation or use, such as applying a rule or principle.
2. It can also refer to the act of making a formal request or submission, such as applying for a job or a loan.
3. In the context of mathematics or computing, “apply” can refer to the act of using a function or operation to a set of values or arguments.
4. In the context of physical sciences, “apply” can refer to the act of exerting force or pressure on an object, such as applying pressure to a wound.
5. In general, “apply” refers to the act of putting something into action or practice, or making use of something for a specific purpose.

CO-4 Determine and calculate various physical properties of a given material by using

1. **Density:** The density of a material is defined as its mass per unit volume. It can be calculated by dividing the mass of the material by its volume.
2. **Elasticity:** Elasticity is the ability of a material to return to its original shape after being deformed. The modulus of elasticity, also known as Young’s modulus, is a measure of the stiffness of a material and can be calculated by dividing the stress by the strain.

3. **Thermal conductivity:** Thermal conductivity is a measure of a material's ability to conduct heat. It can be calculated by dividing the heat flow rate by the temperature gradient.
4. **Electrical conductivity:** Electrical conductivity is a measure of a material's ability to conduct electricity. It can be calculated by dividing the current by the electric field.
5. **Hardness:** Hardness is a measure of a material's resistance to deformation or penetration. It can be determined using various hardness tests, such as the Rockwell, Brinell, or Vickers tests.
6. **Melting point:** The melting point of a material is the temperature at which it changes from a solid to a liquid state. It can be determined using a melting point apparatus or by observing the material as it is heated.
7. **Tensile strength:** Tensile strength is a measure of the maximum stress a material can withstand before breaking. It can be determined by conducting a tensile test on a sample of the material.
8. **Compressive strength:** Compressive strength is a measure of the maximum stress a material can withstand before failing in compression. It can be determined by conducting a compressive test on a sample of the material.

These are some of the physical properties of a material that can be determined and calculated using various methods and techniques. It is important to understand these properties in order to select the appropriate material for a given application.

Various Experiments

There are numerous experiments that have been conducted in various fields of science. Some of the most notable experiments include:

1. **The Double-Slit Experiment:** This experiment, first performed by Thomas Young in 1801, demonstrated the wave-like behavior of light. It involved shining a beam of light through two narrow slits and observing the resulting interference pattern on a screen behind the slits.
2. **The Michelson-Morley Experiment:** This experiment, conducted by Albert Michelson and Edward Morley in 1887, was designed to detect the presence of the hypothetical "luminiferous ether" that was thought to permeate all space and serve as the medium through which light waves traveled. The experiment failed to detect any such ether, providing evidence for the theory of relativity.
3. **The Rutherford Gold Foil Experiment:** This experiment, conducted by Ernest Rutherford in 1911, involved firing alpha particles at a thin sheet of gold foil. The results showed that most of the alpha particles passed straight through the foil, while a small number were deflected at large angles. This led Rutherford to propose the existence of a small, dense nucleus at the center of the atom.

4. **The Stanford Prison Experiment:** This experiment, conducted by Philip Zimbardo in 1971, involved assigning college students to play the roles of prisoners and guards in a simulated prison environment. The experiment had to be terminated early due to the abusive behavior of the guards and the psychological distress experienced by the prisoners.
5. **The Milgram Experiment:** This experiment, conducted by Stanley Milgram in 1961, involved instructing participants to administer electric shocks to a “learner” (who was actually a confederate of the experimenter) whenever the learner made a mistake on a memory task. The results showed that a majority of participants were willing to administer what they believed to be painful and potentially dangerous shocks when instructed to do so by an authority figure.

These are just a few examples of the many experiments that have been conducted in the pursuit of scientific knowledge. Each experiment has contributed to our understanding of the world around us and has helped to shape the course of scientific inquiry.

Apply

1. The word “apply” can have several meanings, depending on the context in which it is used.
2. It can mean to make a formal request, usually in writing, for something such as a job, a loan, or admission to a school or program.
3. It can also mean to put or spread something, such as a substance or a force, onto or over something else.
4. Another meaning of “apply” is to use or be relevant to a particular situation or circumstance.
5. In mathematics, “apply” can mean to use a rule or formula to solve a problem or to calculate a result.

CO-5 Study and estimate the performance and parameter of given equipment by using

1. To study and estimate the performance and parameter of given equipment, it is important to first understand the specifications and capabilities of the equipment.
2. This can be done by referring to the equipment’s manual or technical documentation, which should provide detailed information on its performance and parameters.
3. Once the specifications and capabilities of the equipment are understood, various tests and measurements can be performed to evaluate its performance and determine its parameters.
4. These tests and measurements may include, but are not limited to, power consumption, efficiency, accuracy, and reliability.

5. The results of these tests and measurements can then be compared to the equipment's specifications to determine if it is performing as expected.
6. If the equipment is not performing as expected, further investigation may be necessary to identify and address any issues.
7. It is important to regularly monitor and evaluate the performance and parameters of equipment to ensure that it is operating at its optimal level. This can help to prevent potential issues and improve overall efficiency.

Graphical and Computational Analysis

Graphical and computational analysis are two methods used to analyze data and solve problems. Here are some key points to consider:

1. **Graphical analysis** involves the use of graphs and charts to visually represent data and identify trends or patterns. This method is useful for quickly identifying relationships between variables and making predictions.
2. **Computational analysis** involves the use of mathematical algorithms and computer programs to analyze data and solve problems. This method is useful for handling large datasets and performing complex calculations.
3. Both methods have their strengths and weaknesses, and the choice of method depends on the specific problem at hand. Graphical analysis is often more intuitive and easier to understand, while computational analysis can be more powerful and precise.
4. In many cases, a combination of both methods can be used to achieve the best results. For example, graphical analysis can be used to identify trends and relationships in the data, while computational analysis can be used to perform more detailed calculations and make predictions.
5. It is important to have a solid understanding of both methods in order to effectively analyze data and solve problems. This can involve learning about different types of graphs and charts, as well as different mathematical algorithms and computer programs.

Apply

1. The term “apply” can refer to the act of putting something into operation or use.
2. It can also mean to make a formal request, such as applying for a job or a loan.
3. In the context of mathematics, “apply” can mean to use a mathematical operation or formula to a given set of values.
4. In the context of science, “apply” can mean to use a scientific principle or theory to explain or predict a phenomenon.
5. In general, “apply” can mean to put into practice or to use something in a practical way.

Reference Books

- Reference books are books that contain factual information and are intended to be consulted rather than read from cover to cover.
- They are often used to provide background information or to quickly find specific facts or data.
- Examples of reference books include dictionaries, encyclopedias, almanacs, and atlases.
- Reference books are often organized in a way that makes it easy to find the information you need, such as alphabetically or by subject.
- They are usually written by experts in their field and are considered reliable sources of information.
- Reference books can be found in libraries, bookstores, and online.
- They are an important resource for students, researchers, and anyone seeking accurate and reliable information on a particular topic.

1. Practical Physics- K. K. Dey & B. N. Dutta (Kalyani Publishers New Delhi)

- “Practical Physics” is a book written by K. K. Dey and B. N. Dutta.
- It is published by Kalyani Publishers, New Delhi.
- The book is a comprehensive guide to practical physics experiments and measurements.
- It covers a wide range of topics, including mechanics, heat, optics, electricity, magnetism, and modern physics.
- The book is written in a clear and concise manner, with an emphasis on experimental techniques and data analysis.
- It is a valuable resource for students and teachers of physics, as well as for anyone interested in conducting physics experiments.

2. Engineering Physics-Theory and Practical- Katiyar & Pandey (Wiley India)

- “Engineering Physics-Theory and Practical” is a book written by Katiyar & Pandey and published by Wiley India.
- The book covers the fundamental concepts of physics and their applications in engineering.
- It is designed to provide a comprehensive understanding of the subject for students pursuing engineering courses.
- The book is divided into two parts: theory and practical.
- The theory part covers topics such as mechanics, waves and oscillations, optics, electromagnetism, and modern physics.
- The practical part includes experiments and activities that help students understand the concepts discussed in the theory part.
- The book is written in a clear and concise manner, making it easy for students to understand the subject.

- It includes numerous examples, illustrations, and exercises to help students learn and practice the concepts.
- Overall, “Engineering Physics-Theory and Practical” by Katiyar & Pandey is a valuable resource for students studying engineering physics.

3. Engineering Physics Practical- S K Gupta (Krishna Prakashan Meerut)

- Engineering Physics Practical is a book written by S K Gupta and published by Krishna Prakashan Meerut.
- The book is intended for students studying engineering and covers practical aspects of physics.
- It includes experiments and practical exercises to help students understand the concepts of physics and apply them in an engineering context.
- The book is written in a clear and concise manner, making it easy for students to follow and understand.
- It is a valuable resource for students preparing for exams or for anyone looking to improve their understanding of physics in an engineering context.